

SOUTHERN ILLINOIS, IL, USA

WMO: 724336

Lat: 37.780N

Lon: 89.250W

Elev: 406

StdP: 14.48

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 93810

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.6	14.4	-0.5	5.4	11.7	3.8	6.8	16.1	27.5	50.5	25.3	49.1	6.8	330	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	93.6	76.9	91.4	76.5	89.8	75.8	80.1	89.6	78.6	88.1	77.5	86.6	8.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	144.4	85.7	75.6	136.1	84.0	74.7	131.9	82.9	43.8	88.9	42.3	88.7	41.1	86.5	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.8	19.9	17.5	DB	1.6	97.5	5.2	3.2	-2.1	99.8	-5.2	101.7	-8.1	103.5	-11.9	105.9	
			WB	1.2	82.2	5.4	1.2	-2.6	83.0	-5.8	83.7	-8.8	84.4	-12.7	85.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.1	33.8	37.6	46.7	57.5	67.1	75.4	78.1	76.5	69.3	58.0	46.5	37.1
	DBStd	17.90	11.01	10.88	10.54	8.76	7.61	5.58	4.99	4.90	6.90	8.98	9.63	10.41
	HDD50	1732	517	368	191	31	1	0	0	0	0	26	179	419
	HDD65	4353	969	768	572	251	71	4	0	1	37	256	558	866
	CDD50	4304	13	20	88	257	529	761	872	822	579	273	72	18
	CDD65	1452	0	0	4	27	135	315	407	358	166	38	2	0
	CDH74	14154	0	0	23	205	1143	3160	4198	3417	1622	371	15	0
	CDH80	5622	0	0	0	22	316	1295	1839	1455	602	93	0	0
Wind	WSAvg	7.2	8.8	8.9	9.0	9.2	7.3	6.3	5.2	4.6	4.7	6.4	7.8	8.1
Precipitation	PrecAvg	45.8	3.1	2.8	4.2	5.0	5.2	4.3	3.8	3.1	3.3	3.2	4.3	3.5
	PrecMax	66.0	6.6	6.1	11.5	15.0	9.9	9.6	7.2	7.6	9.6	8.0	8.5	8.1
	PrecMin	30.7	0.6	0.6	1.8	1.7	1.3	1.3	1.5	0.6	0.6	0.6	0.4	0.7
	PrecStd	8.2	1.5	1.3	2.1	2.8	2.4	2.2	1.8	1.5	1.9	1.9	2.3	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.6	70.2	77.4	82.7	89.7	94.6	99.2	97.7	93.3	87.3	76.2	67.1
		MCWB	60.9	61.7	63.8	66.3	73.6	75.8	76.9	76.9	72.9	72.0	63.3	62.1
	2%	DB	62.0	65.0	72.8	79.5	86.4	92.0	93.5	93.2	90.0	82.2	70.8	62.9
		MCWB	58.3	57.7	62.0	65.5	72.3	75.9	78.1	77.0	73.9	68.4	60.9	58.2
	5%	DB	56.6	60.7	68.7	75.9	83.7	90.2	91.2	90.5	87.0	78.3	66.4	57.7
		MCWB	51.9	54.3	59.1	63.8	70.8	75.1	77.9	76.3	72.9	65.7	58.6	53.2
	10%	DB	51.4	55.3	64.2	72.9	81.0	87.9	89.3	88.0	83.2	73.7	63.3	53.1
		MCWB	46.4	48.6	56.2	62.0	69.3	74.1	77.0	75.3	70.4	63.5	56.2	48.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.1	63.6	66.4	70.5	76.3	80.2	82.3	81.6	77.9	74.0	67.0	63.5
		MCDB	64.4	68.2	74.4	79.4	85.3	89.6	91.3	90.7	87.1	83.6	72.9	67.2
	2%	WB	58.9	59.5	63.7	68.0	74.4	78.3	80.6	79.4	76.2	71.4	63.3	58.9
		MCDB	62.2	63.4	71.2	76.1	83.5	88.6	89.9	89.4	85.4	79.1	67.3	62.3
	5%	WB	53.0	55.2	60.7	66.0	72.7	77.0	79.2	78.0	74.6	68.8	60.3	54.3
		MCDB	55.4	59.4	67.2	73.5	81.0	86.9	88.7	87.6	83.2	74.9	65.9	57.3
	10%	WB	46.1	49.2	57.6	63.9	70.9	75.6	78.0	76.7	73.0	65.7	56.6	48.6
		MCDB	50.2	54.1	63.1	70.4	78.5	84.8	86.6	85.5	80.3	71.9	62.4	52.3
Mean Daily Temperature Range	5% DB	MDBR	16.5	17.3	19.2	20.8	19.9	20.0	19.1	20.8	23.5	23.0	20.1	16.0
		MCDBR	21.3	24.2	23.5	23.7	22.4	22.1	20.9	23.8	26.6	26.6	24.7	20.4
		MCWBR	17.9	18.6	15.2	13.2	10.8	9.0	8.6	10.1	11.9	14.0	16.5	16.7
	5% WB	MCDBR	18.7	22.1	20.9	19.7	19.4	19.3	18.8	20.5	21.5	22.2	20.7	18.0
		MCWBR	17.6	18.7	14.7	12.6	10.5	9.1	9.0	9.8	10.8	12.9	16.4	16.7
Clear-Sky Solar Irradiance	taub	0.314	0.325	0.363	0.395	0.429	0.438	0.473	0.468	0.408	0.357	0.320	0.306	
	taud	2.466	2.418	2.372	2.281	2.242	2.232	2.156	2.162	2.320	2.454	2.499	2.493	
	Ebn at Noon	276	287	285	280	272	268	258	256	266	272	272	271	
	Edh at Noon	26	31	36	42	44	44	48	46	37	30	25	24	
All-Sky Solar Radiation	RadAvg	665	878	1201	1583	1784	2042	1969	1797	1502	1116	754	561	
	RadStd	68	104	110	102	154	128	126	97	93	116	83	52	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139

Nomenclature: See separate page